

Peijing Li

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EDUCATION

Stanford University

Stanford, CA

Doctor of Philosophy in Electrical Engineering

Sep 2024 – Present

- Coursework: Engineering Design Optimization, Formal Methods for Computer Systems, Emerging Non-Volatile Memory Devices and Circuit Design, Computer Systems Architecture, Parallel Computing

University of Michigan

Ann Arbor, MI

Bachelor of Science in Engineering in Computer Science

Aug 2020 – May 2024

Minor in Civil Engineering; College of Engineering Honors Program

- Cumulative GPA: 4.0/4.0. Graduated *summa cum laude*
- Coursework: Computer Architecture, Advanced Compilers, Computer Security, Computer Vision, Software Engineering, Transportation Engineering, Linear Optimization

RESEARCH EXPERIENCE

Stanford Differentiated Access Memory Project

Sep 2024 – Present

- Designing architectural and compiler support for heterogeneous on-chip memory systems composed short- and long-term memory devices in state-of-the-art AI accelerator hardware
- Leading tapeout project on 3T gain cell eDRAM-based AI accelerator, implementing high-level synthesis design of dynamically refreshed gain cell on-chip memory and retention-aware accelerator processing element cores
- Developing GainSight, a profiler suite that measures the fine-grained data lifetimes and memory access patterns of software applications and aligns these patterns to guide the composition of heterogeneous memory systems

WORK EXPERIENCE

Stanford University Department of Computer Science

Stanford, CA

Teaching Assistant, CS 217 Hardware Accelerators for Machine Learning

Jan 2026 – Mar 2026

ASML Silicon Valley

San José, CA

Software Test Engineer Intern

May 2023 – Aug 2023

University of Michigan Department of Civil and Environmental Engineering

Ann Arbor, MI

Instructional Aide, CEE 375 Sensors, Circuits, and Signals

Jan 2023 – Apr 2023

University of Michigan Transportation Research Institute

Ann Arbor, MI

Research Assistant, Biosciences Group

Feb 2022 – Aug 2022

Dell Technologies

Shanghai, China

Software Engineering Intern, Research & Development

Jun 2021 – Aug 2021

PUBLICATIONS

[1] Peijing Li, Muhammad Shahir Abdurrahman, Rachel Cleaveland, Sergey Legtchenko, Philip Levis, Ioan Stefanovici, Thierry Tambe, David Tennenhouse, Caroline Trippel, and H.-S. Philip Wong. 2025. Towards Memory Specialization: A Case for Long-Term and Short-Term RAM. In Workshop on Disruptive Memory Systems (DIMES '25), October 13, 2025. Association for Computing Machinery, Seoul, Korea (South), 10.

<https://doi.org/10.1145/3764862.3768175>

[2] Peijing Li, Matthew Hung, Yiming Tan, Konstantin Hoßfeld, Jake Cheng Jiajun, Shuhan Liu, Lixian Yan, Xinxin Wang, Philip Levis, H.-S. Philip Wong, and Thierry Tambe. 2025. GainSight: A Unified Framework for Data Lifetime Profiling and Heterogeneous Memory Composition.

<https://doi.org/10.48550/arXiv.2504.14866>

[3] Samuel Dayo, Shuhan Liu, Peijing Li, Philip Levis, Subhasish Mitra, Thierry Tambe, David Tennenhouse, and H.-S. Philip Wong. 2025. The Future of Memory: Limits and Opportunities. <https://doi.org/10.48550/arXiv.2508.20425>

[4] Xinxin Wang, Lixian Yan, Shuhan Liu, Luke Upton, Zhuoqi Cai, Yiming Tan, Shengman Li, Koustav Jana, Peijing Li, Jesse Cirimelli-Low, Thierry Tambe, Matthew Guthaus, and H.-S. Philip Wong. 2025. OpenGCRAM: An Open-Source Gain Cell Compiler Enabling Design-Space Exploration for AI Workloads. <https://doi.org/10.48550/arXiv.2507.10849>